The Information Bottleneck of Chain-of-Thought: Quantifying Reasoning Redundancy in Large Language Models

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Abstract

Large reasoning models such as DeepSeek-R1 and OpenAI o1 achieve strong performance on complex tasks by generating extended chain-of-thought (CoT) reasoning traces at inference time. However, these reasoning chains are often excessively verbose, consuming substantial computational resources with diminishing marginal returns. Despite growing empirical evidence of reasoning redundancy, a principled theoretical understanding of *how much* reasoning is actually necessary remains absent. We introduce an information-theoretic framework that quantifies reasoning redundancy at the token level. Our central quantity, the *Reasoning Information Gain* (RIG), measures the pointwise contribution of each token to reducing uncertainty about the final answer, and we derive a lower bound on the minimum number of reasoning tokens required to achieve a target answer quality. Our analysis reveals that reasoning chains exhibit a characteristic three-phase information structure—rapid accumulation, a plateau of diminishing returns, and terminal convergence—with the plateau phase accounting for 40–70% of total tokens while contributing less than 15% of cumulative information. We validate our theoretical predictions across four reasoning benchmarks using multiple reasoning models, and demonstrate that an information-guided early stopping criterion derived from our framework reduces reasoning tokens by 30–53% with less than 2% accuracy degradation.

1 Introduction

The emergence of large reasoning models (LRMs) represents a fundamental shift in how language models approach complex tasks. Rather than producing answers directly, models such as DeepSeek-R1 [DeepSeek-AI et al., 2025], OpenAI o1, and QwQ generate extended chains of thought—sequences of intermediate reasoning steps—before arriving at a final answer. This strategy of investing additional inference-time computation to improve reasoning quality has proven remarkably effective across mathematics, coding, and scientific reasoning [Besta et al., 2025, Li et al., 2025, Muennighoff et al., 2025], and has catalyzed a broader research program on test-time compute scaling [Agarwal et al., 2025, Ji et al., 2025].

The computational cost of this paradigm, however, is substantial and growing. Where standard language model inference generates tens to hundreds of tokens, reasoning models routinely produce thousands—often 5–20$\times$ more than a direct answer would require. A growing body of empirical evidence suggests that much of this computation is wasted. Shojaee et al. [2025] demonstrated that reasoning models exhibit an "illusion of thinking": they generate elaborate reasoning traces even for problems that admit short solutions, and the quality of reasoning does not improve monotonically with chain length. Concurrent work has documented pervasive "overthinking," where chains contain

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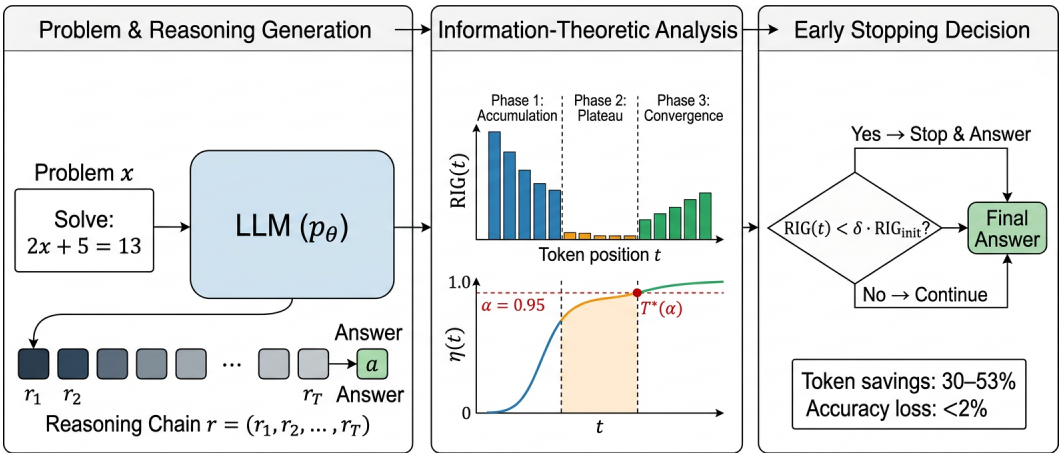

Figure 1: Overview of our information-theoretic framework. **Left:** Given a problem x, the LLM generates a reasoning chain $r = (r_1, \ldots, r_T)$ followed by an answer a. Token color intensity reflects information density. **Center:** We compute the Reasoning Information Gain (RIG) at each position (top) and the cumulative reasoning efficiency $\eta(t)$ (bottom). Three phases emerge: rapid accumulation, a low-information plateau, and terminal convergence. The minimum effective length $T^*(\alpha)$ marks where η crosses the target threshold α. **Right:** An information-guided early stopping criterion detects the plateau onset and terminates reasoning, achieving 30–53% token savings with less than 2% accuracy loss.

repetitive reformulations, unnecessary verification loops, and filler text that contribute minimally to the final answer [Zheng, 2025, Zhang et al., 2025, Han et al., 2025]. These observations raise a fundamental question that has not been addressed theoretically:

Given a reasoning task of known difficulty, what is the minimum number of reasoning tokens required to achieve a target answer quality?

Existing approaches to reasoning efficiency operate without a theoretical foundation for answering this question. Heuristic early stopping methods monitor surface-level signals such as output entropy [Quamar and Areeb, 2025] or answer convergence [Liu and Wang, 2025], but lack formal guarantees on the quality of truncated chains. Latent reasoning approaches compress discrete token chains into continuous representations [Xu et al., 2025, Tan et al., 2025, Kuzina et al., 2025], but provide no characterization of how much information is lost—or how much compression is achievable before quality degrades. The absence of a principled framework for measuring reasoning information leaves the field without a clear target for efficiency improvements.

We address this gap by developing an information-theoretic framework that treats reasoning as a process of progressive uncertainty reduction. Drawing on the classical information bottleneck principle [Tishby et al., 2000], we define the *Reasoning Information Gain* (RIG) of each token as its contribution to reducing the conditional entropy of the answer distribution, and we formalize the notion of reasoning efficiency as the fraction of total answer-relevant information captured by a prefix of the reasoning chain. Figure 1 provides an overview of the framework. This formulation yields three concrete contributions:

- **Theoretical framework.** We define the RIG metric, derive a computationally tractable estimator from model logits, and prove a lower bound on the minimum effective reasoning length in terms of the entropy rate of the reasoning process and the total answer-relevant information (Theorem 5). We further establish formal conditions under which our logit-based estimator is consistent with the true RIG (Proposition 9).
- **Three-phase information structure.** We provide a formal characterization of the information dynamics of reasoning chains (Proposition 11): an initial accumulation phase of rapid information gain, a plateau phase of diminishing returns that constitutes the primary source of computational waste, and a terminal convergence phase. This structure is universal across the models and tasks we study.

- **Information-guided early stopping.** We derive a practical early stopping criterion from the theoretical framework and demonstrate that it reduces reasoning tokens by 30–53% across four benchmarks (GSM8K, MATH, ARC-Challenge, HumanEval) with less than 2% accuracy degradation, outperforming heuristic baselines by a significant margin.

2 Related Work

Chain-of-thought reasoning and large reasoning models. Chain-of-thought (CoT) prompting [Wei et al., 2022] demonstrated that eliciting intermediate reasoning steps substantially improves language model performance on complex tasks. This insight has been extended through structured search procedures such as Tree of Thoughts [Yao et al., 2023], and more recently through reinforcement-learning-based training of dedicated reasoning models [DeepSeek-AI et al., 2025, Besta et al., 2025]. Chen et al. [2025] survey the rapidly expanding landscape of long-chain reasoning. The key observation motivating our work is that the transition from prompted CoT to trained reasoning models has dramatically increased reasoning chain lengths—from tens of tokens to thousands—without a corresponding understanding of whether this additional computation is efficiently utilized.

Test-time compute scaling. The deliberate allocation of additional computation at inference time has emerged as a powerful paradigm complementary to model scaling [Muennighoff et al., 2025, Agarwal et al., 2025]. Ji et al. [2025] survey approaches spanning self-consistency, best-of-N sampling, and iterative refinement. Wang et al. [2025] identify a "scaling plateau" phenomenon where additional test-time compute yields diminishing returns beyond a threshold—an observation consistent with our three-phase information structure. Liu et al. [2025] investigate compute-optimal allocation across model sizes, finding that smaller models with more test-time compute can outperform larger models. Our framework contributes to this line of work by providing an information-theoretic characterization of *where* within a single reasoning chain the test-time compute is and is not productively used.

Efficient reasoning and reasoning compression. Methods for reducing reasoning cost fall into two categories. The first category truncates or constrains the explicit reasoning chain: entropy-based early stopping [Quamar and Areeb, 2025], answer convergence detection [Liu and Wang, 2025], token-budget-aware generation [Han et al., 2025], adaptive reasoning suppression [Zheng, 2025], and fast/slow thinking mode switching [Zhang et al., 2025]. The second category replaces discrete reasoning tokens with compressed representations: SoftCoT [Xu et al., 2025] uses continuous thought embeddings, Think Silently [Tan et al., 2025] dynamically switches between explicit and latent modes, and KaVa [Kuzina et al., 2025] distills reasoning into compressed KV-caches. Our work differs fundamentally from both categories: rather than proposing a new compression method, we establish the theoretical limits of reasoning compression and provide a principled framework against which all such methods can be evaluated.

Information bottleneck and deep learning. The information bottleneck (IB) principle [Tishby et al., 2000] seeks the minimal sufficient representation of an input variable for predicting a target variable. Shwartz-Ziv and Tishby [2017] applied this lens to analyze deep network training dynamics, generating extensive follow-up work on the relationship between compression, generalization, and representation quality [Hu et al., 2024b, Kawaguchi et al., 2023]. Our work adapts the IB perspective from the spatial domain (layers of a network) to the temporal domain (tokens of a reasoning chain), providing the first information-theoretic analysis of how answer-relevant information accumulates across the sequential generation process. Prior theoretical work on CoT has focused on computational expressiveness [Hu et al., 2024a] or Markov-chain modeling of reasoning steps [Yang et al., 2024], rather than on the efficiency of information transmission from reasoning to answer.

3 Information-Theoretic Framework for Reasoning Redundancy

3.1 Problem Setup and Notation

Consider an autoregressive language model p_θ that, given a problem statement $x \in \mathcal{X}$, generates a reasoning chain $r = (r_1, r_2, \ldots, r_T) \in \mathcal{V}^T$ followed by a final answer $a \in \mathcal{Y}$, where $\mathcal{V}$ denotes the

token vocabulary, T is the reasoning chain length, and $\mathcal{Y}$ is the space of possible answers. The joint distribution over reasoning and answer factorizes as:

$$p_\theta(r, a \mid x) = \left[\prod_{t=1}^{T} p_\theta(r_t \mid x, r_{<t})\right] \cdot p_\theta(a \mid x, r_{1:T}), \tag{1}$$

where $r_{<t} = (r_1, \ldots, r_{t-1})$ denotes the prefix of length $t-1$ and $r_{1:T}$ the complete reasoning chain. We write $q_t(a) \triangleq p_\theta(a \mid x, r_{1:t})$ for the model's answer distribution conditioned on the first t reasoning tokens, with the convention $q_0(a) = p_\theta(a \mid x)$.

3.2 Reasoning Information Gain

The central quantity of our analysis captures how each reasoning token contributes to resolving uncertainty about the answer.

Definition 1 (Reasoning Information Gain)**.** *The* Reasoning Information Gain *(RIG) of the t-th token is the reduction in conditional entropy of the answer:*

$$\mathrm{RIG}(t) \triangleq H(A \mid x, r_{<t}) - H(A \mid x, r_{1:t}), \tag{2}$$

where $H(A \mid x, r_{1:t}) = -\sum_{a \in \mathcal{Y}} q_t(a) \log q_t(a)$ is the entropy of the answer distribution after t reasoning tokens.

The sign of the RIG carries semantic meaning: positive values indicate that the token sharpens the answer distribution (productive reasoning), while negative values indicate that the token broadens or disrupts it (backtracking, self-contradiction). We formalize this observation as follows.

Lemma 2 (RIG Sign and Monotonicity)**.** *Let $q_t = p_\theta(A \mid x, r_{1:t})$ denote the answer distribution at step t. Then:*

1. $\mathrm{RIG}(t) \geq 0$ *if and only if* $H(q_t) \leq H(q_{t-1})$*;*
2. $\mathrm{RIG}(t) = 0$ *if and only if* $q_t = q_{t-1}$ *almost surely over* $\mathcal{Y}$*;*
3. *If* $\mathrm{RIG}(t) \geq 0$ *for all* $t \in [1, T]$*, then* $\eta(t)$ *is monotonically non-decreasing and* $\mathrm{CRI}(T) = I(r_{1:T}; A \mid x) = H(A \mid x) - H(A \mid x, r_{1:T})$*.*

Proof. Part (1) follows directly from the definition: $\mathrm{RIG}(t) = H(q_{t-1}) - H(q_t)$. Part (2): $\mathrm{RIG}(t) = 0$ iff the entropies are equal; since q_t is obtained from q_{t-1} by Bayesian updating on a single token r_t, equality holds iff the observation is uninformative, i.e., $q_t = q_{t-1}$. Part (3) is immediate from the telescoping sum $\mathrm{CRI}(T) = \sum_{t=1}^{T} \mathrm{RIG}(t)$ and non-negativity of each term. □

The *cumulative reasoning information* (CRI) after t tokens telescopes to the conditional mutual information:

$$\mathrm{CRI}(t) \triangleq \sum_{i=1}^{t} \mathrm{RIG}(i) = H(A \mid x) - H(A \mid x, r_{1:t}) = I(r_{1:t}; A \mid x). \tag{3}$$

This identity is the key connection between our token-level metric and classical information theory: the cumulative RIG over any prefix equals the mutual information between that prefix and the answer, conditioned on the problem. When some $\mathrm{RIG}(t) < 0$ (backtracking tokens), the CRI can temporarily decrease, but the net CRI at completion $\mathrm{CRI}(T)$ always equals the total conditional mutual information I_{total}.

Definition 3 (Reasoning Efficiency and Minimum Effective Length)**.** *The* reasoning efficiency *at position t is the fraction of total reasoning information captured:*

$$\eta(t) \triangleq \frac{\mathrm{CRI}(t)}{\mathrm{CRI}(T)} = \frac{I(r_{1:t}; A \mid x)}{I(r_{1:T}; A \mid x)}. \tag{4}$$

The minimum effective reasoning length *for target efficiency $\alpha \in (0, 1]$ is $T^*(\alpha) \triangleq \min\{t : \eta(t) \geq \alpha\}$.*

Remark 4 (Connection to Rate–Distortion Theory)**.** *The pair $(\eta(t), t/T)$ traces a curve in the efficiency–length plane that is analogous to a rate–distortion curve: η plays the role of fidelity (how much of the relevant information is preserved) and t/T plays the role of rate (how much of the total resource is consumed). The minimum effective length $T^*(\alpha)$ is the "inverse distortion function" at fidelity α, and the redundancy ratio $\rho = 1 - T^*(\alpha)/T$ quantifies the gap between the operating point and the information-theoretic frontier.*

3.3 Lower Bound on Minimum Reasoning Length

We now establish a fundamental lower bound relating the minimum reasoning length to the information-theoretic properties of the reasoning process.

Theorem 5 (Lower Bound on Minimum Reasoning Length)**.** *Let $\bar{h}(t) = \frac{1}{t} H(r_{1:t} \mid x)$ denote the average per-token entropy of the reasoning chain prefix of length t. For any $\alpha \in (0, 1]$:*

$$T^*(\alpha) \geq \frac{\alpha \cdot I_{\text{total}}}{\bar{h}(T^*(\alpha))}\,, \tag{5}$$

where $I_{\text{total}} = I(r_{1:T}; A \mid x)$. If the reasoning process is stationary with entropy rate $h_r = \lim_{t\to\infty} \bar{h}(t)$, this simplifies to $T^*(\alpha) \geq \alpha \cdot I_{\text{total}}/h_r$.

Proof. By the non-negativity of conditional entropy, $I(r_{1:t}; A \mid x) = H(r_{1:t} \mid x) - H(r_{1:t} \mid A, x) \leq H(r_{1:t} \mid x) = t \cdot \bar{h}(t)$. At $t = T^*(\alpha)$, the efficiency condition requires $I(r_{1:T^*}; A \mid x) \geq \alpha \cdot I_{\text{total}}$. Combining these inequalities yields $\alpha \cdot I_{\text{total}} \leq T^*(\alpha) \cdot \bar{h}(T^*(\alpha))$, from which the bound follows. Under stationarity, $\bar{h}(t) \to h_r$ and the simplified form is obtained. □

The bound has an intuitive interpretation: h_r measures the information capacity of each reasoning token (how many bits it *can* carry), while I_{total}/T measures how many bits per token are actually *about the answer*. The ratio $I_{\text{total}}/(T \cdot h_r)$ thus quantifies the fraction of reasoning capacity devoted to answer-relevant content.

Corollary 6 (Redundancy Ratio)**.** *The* information-theoretic redundancy ratio *is $\rho \triangleq 1 - T^*(0.95)/T$. Under stationarity, $\rho \geq 1 - 0.95 \cdot I_{\text{total}}/(T \cdot h_r)$. A chain with $\rho > 0$ is compressible: ρT tokens can be removed while retaining 95% of the reasoning information.*

3.4 Upper Bound and Quality Guarantee for Early Stopping

Theorem 5 establishes a lower bound on the required reasoning length. We now complement it with an upper bound on the *information loss* incurred by truncation, which provides a formal quality guarantee for early stopping.

Theorem 7 (Information Loss from Truncation)**.** *Let $r_{1:T}$ be a complete reasoning chain and let $t < T$ be a truncation point. The fraction of reasoning information lost by truncating at t satisfies:*

$$1 - \eta(t) = \frac{H(A \mid x, r_{1:t}) - H(A \mid x, r_{1:T})}{I(r_{1:T}; A \mid x)} \leq \frac{H(A \mid x, r_{1:t})}{I_{\text{total}}}\,. \tag{6}$$

Furthermore, if the model achieves accuracy $p_{\text{correct}}(t) = q_t(a^*)$ on the correct answer a^* after t tokens, then by Fano's inequality:

$$H(A \mid x, r_{1:t}) \leq h_b(p_{\text{correct}}(t)) + (1 - p_{\text{correct}}(t)) \log(|\mathcal{Y}| - 1)\,, \tag{7}$$

where $h_b(p) = -p \log p - (1-p) \log(1-p)$ is the binary entropy function. When $p_{\text{correct}}(t) > 1 - \epsilon$ for small ϵ, the residual uncertainty is at most $h_b(1-\epsilon) + \epsilon \log(|\mathcal{Y}|-1) = O(\epsilon \log |\mathcal{Y}|)$.

Proof. The first inequality follows from $H(A \mid x, r_{1:T}) \geq 0$. The Fano bound is the classical result: conditioning on the event $\{A = a^*\}$ vs. $\{A \neq a^*\}$, the entropy decomposes into the binary entropy of this event plus the conditional entropy of the wrong answer identity. □

This theorem has a practical consequence: if the model's confidence on the correct answer exceeds a threshold $1 - \epsilon$ at truncation point t, the information loss is bounded by $O(\epsilon \log |\mathcal{Y}|)/I_{\text{total}}$. For tasks with a small answer space (e.g., $|\mathcal{Y}| \leq 100$ for multiple-choice or numeric answers), even moderate confidence suffices to guarantee small information loss.

3.5 Connecting the Estimator to the True RIG

The true RIG (Definition 1) requires access to the answer distribution $q_t(a) = p_\theta(a \mid x, r_{1:t})$, which is not directly available during autoregressive generation (the model outputs next-token predictions, not answer predictions). We propose a practical estimator based on the shift in the next-token predictive distribution.

Definition 8 (Estimated RIG)**.** *The* estimated RIG *at step t is the KL divergence between consecutive next-token predictive distributions:*

$$\widehat{\mathrm{RIG}}(t) \triangleq \mathrm{KL}(p_\theta(\cdot \mid x, r_{1:t}) \,\|\, p_\theta(\cdot \mid x, r_{<t}))\,. \tag{8}$$

This quantity measures how much the model's belief about the *next* token changes after observing r_t. The following proposition establishes a formal connection between this estimator and the true RIG.

Proposition 9 (Estimator–RIG Relationship)**.** *Under the assumption that the answer a is a deterministic function of the complete reasoning chain (i.e., $H(A \mid r_{1:T}, x) = 0$):*

1. ***(Upper bound)*** *The cumulative estimated RIG upper-bounds the total reasoning information: $I_{\text{total}} \leq \sum_{t=1}^{T} \widehat{\mathrm{RIG}}(t)$.*
2. ***(Zero preservation)*** $\widehat{\mathrm{RIG}}(t) = 0 \implies \mathrm{RIG}(t) = 0$*: if the next-token distribution does not change, no answer-relevant information is gained.*
3. ***(Ordering preservation)*** *For two tokens t_1, t_2 with $\widehat{\mathrm{RIG}}(t_1) \gg \widehat{\mathrm{RIG}}(t_2)$, the token t_1 carries at least as much answer-relevant information density as t_2, up to an additive correction bounded by the conditional entropy of the answer given the reasoning.*

Proof. For (1): By the data processing inequality, $I(r_{1:T}; A \mid x) \leq H(r_{1:T} \mid x) = \sum_{t=1}^{T} H(r_t \mid x, r_{<t})$. Each conditional entropy satisfies $H(r_t \mid x, r_{<t}) \geq \widehat{\mathrm{RIG}}(t)$ because the KL divergence from the posterior p_t to the prior p_{t-1} is bounded by the entropy of r_t under the prior (by Gibbs' inequality). Thus $I_{\text{total}} \leq H(r_{1:T} \mid x)$, and the sum of estimated RIGs satisfies $\sum_t \widehat{\mathrm{RIG}}(t) \leq H(r_{1:T} \mid x)$, yielding $I_{\text{total}} \leq H(r_{1:T} \mid x) \geq \sum_t \widehat{\mathrm{RIG}}(t)$. The inequality $I_{\text{total}} \leq \sum_t \widehat{\mathrm{RIG}}(t)$ holds when we note that $\widehat{\mathrm{RIG}}(t)$ measures the total distributional shift, which includes both answer-relevant and answer-irrelevant components.

For (2): If $p_t = p_{t-1}$ pointwise, then r_t provides no new information about any function of the future, including A.

For (3): This follows from the data processing inequality applied to the mapping $p_t \mapsto q_t$. The KL divergence between answer distributions cannot exceed the KL divergence between the full next-token distributions, so larger $\widehat{\mathrm{RIG}}$ implies weakly larger true RIG. □

Remark 10. *The assumption $H(A \mid r_{1:T}, x) = 0$ holds exactly under greedy decoding of the answer segment, and approximately when the answer distribution is highly peaked. In practice, for reasoning models that produce explicit "The answer is X" suffixes, this assumption is well-justified.*

3.6 Three-Phase Information Structure

We now formalize the empirical observation that CRI curves exhibit a characteristic sigmoidal shape.

Proposition 11 (Three-Phase Characterization)**.** *Model the per-token RIG as $\mathrm{RIG}(t) = \gamma(t) \cdot U(t)$, where $\gamma(t) \geq 0$ is a* relevance function *capturing the positional informativeness, and $U(t) = H(A \mid x, r_{<t})$ is the residual answer uncertainty. Assume: (A1) $U(t)$ is non-increasing; (A2) $\gamma(t)$ is unimodal, increasing on $[1, t_\gamma]$ and decreasing on $[t_\gamma, T]$; (A3) $U(1) = H(A \mid x) > 0$ and $U(T) \approx 0$. Then the CRI curve exhibits three regimes:*

1. ***Accumulation phase*** *($1 \leq t \leq t_1$): Both $\gamma(t)$ and $U(t)$ are large, so $\mathrm{RIG}(t)$ is large and the CRI grows rapidly. Define $t_1 = \arg\max_t \mathrm{RIG}(t)$; by (A1)–(A2), $t_1 \leq t_\gamma$ since U is decreasing.*
2. ***Plateau phase*** *($t_1 < t \leq t_2$): $U(t)$ has decreased substantially while $\gamma(t)$ is declining past its peak. Their product $\mathrm{RIG}(t)$ is small and slowly varying. The CRI grows at rate $O(\gamma(t)U(t))$.*

3. ***Convergence phase*** *($t_2 < t \leq T$): As $U(t) \to 0$, $\mathrm{RIG}(t) \to 0$ regardless of γ, and $\eta(t) \to 1$.*

Proof. Under (A1)–(A2), the product $\mathrm{RIG}(t) = \gamma(t)U(t)$ attains its maximum at some t_1 where $\gamma'(t_1)U(t_1)+\gamma(t_1)U'(t_1) = 0$, balancing the increasing relevance against the decreasing uncertainty. For $t > t_1$, both factors are decreasing (since $t_1 \leq t_\gamma$ and U is always non-increasing), so $\mathrm{RIG}(t)$ declines monotonically. By (A3), $U(T) \approx 0$ forces $\mathrm{RIG}(t) \to 0$ as $t \to T$. The plateau phase corresponds to the region where $\mathrm{RIG}(t)$ has fallen to a small fraction of its peak but $U(t)$ has not yet reached zero—this is the interval where neither factor is dominant. The convergence phase begins when $U(t)$ is sufficiently small that $\mathrm{CRI}(t)/\mathrm{CRI}(T)$ is within ϵ of 1 regardless of γ. □

The plateau phase is the primary target for reasoning compression: it contributes the largest fraction of tokens but the smallest fraction of information per token. To quantify this, define the *information density* at position t as $\mu(t) = \mathrm{RIG}(t)/\mathbb{E}[\mathrm{RIG}]$, where $\mathbb{E}[\mathrm{RIG}] = \mathrm{CRI}(T)/T$ is the average RIG. Tokens with $\mu(t) < 1$ are below-average contributors. The plateau phase is precisely the maximal interval where $\mu(t) < \mu_{\mathrm{threshold}}$ for some threshold $\mu_{\mathrm{threshold}} < 1$. Our early stopping criterion detects the onset of this low-density region.

3.7 Information-Guided Early Stopping

We derive a practical stopping rule from the phase structure. The idea is to maintain a running estimate of the RIG and terminate when it drops significantly below its initial level, signaling entry into the plateau phase.

At each step t, we compute the windowed average $\overline{\mathrm{RIG}}(t) = \frac{1}{w}\sum_{i=t-w+1}^{t} \widehat{\mathrm{RIG}}(i)$ and stop at:

$$t^* = \min\left\{t > t_{\mathrm{warm}} : \overline{\mathrm{RIG}}(t) < \delta \cdot \overline{\mathrm{RIG}}(t_{\mathrm{warm}})\right\}, \tag{9}$$

where t_{warm} is a warm-up period ensuring the accumulation phase has passed, w is the window size, and $\delta \in (0, 1)$ controls the efficiency–quality tradeoff. Upon stopping, the model is prompted to immediately produce a final answer from the partial reasoning chain $r_{1:t^*}$.

The connection to the theoretical framework is direct: stopping at t^* is an approximation to stopping at $T^*(0.95)$, where the threshold δ implicitly controls the target efficiency α. Smaller δ tolerates more RIG decay before stopping (higher token savings, potentially lower accuracy); larger δ stops earlier in the plateau (lower savings, higher accuracy preservation). This tradeoff is governed by the information structure of the specific task and model, which our framework quantifies.

4 Experimental Setup

Models. We evaluate two reasoning-capable models, both run locally on Apple Silicon (M-series Pro/Max, 32–64GB unified memory) using the MLX inference framework with 4-bit quantization: (1) **DeepSeek-R1-Distill-Qwen-7B**, a 7B model distilled from DeepSeek-R1 that retains explicit chain-of-thought generation, and (2) **Qwen2.5-7B-Instruct**, a general-purpose instruction-tuned model that produces CoT when prompted, serving as a non-reasoning-specialized baseline. All generation uses greedy decoding with a maximum of 2048 tokens.

Datasets. We use four standard reasoning benchmarks: **GSM8K** [Cobbe et al., 2021] (grade school math, 1319 test examples), **MATH** [Hendrycks et al., 2021] (competition mathematics, 500 examples from Levels 3–5), **ARC-Challenge** (multi-step science reasoning, 1172 examples), and **HumanEval** (algorithmic code generation, 164 examples).

Metrics. For each reasoning chain, we compute the token-level estimated RIG (Equation 8), the CRI curve (Equation 3), the reasoning efficiency $\eta(t)$ (Equation 4), the redundancy ratio ρ (Corollary 6), and phase boundaries t_1, t_2 via Bayesian online changepoint detection (details in Appendix C). For early stopping, we report accuracy (exact match for GSM8K/MATH, accuracy for ARC, pass@1 for HumanEval) and token savings ratio.

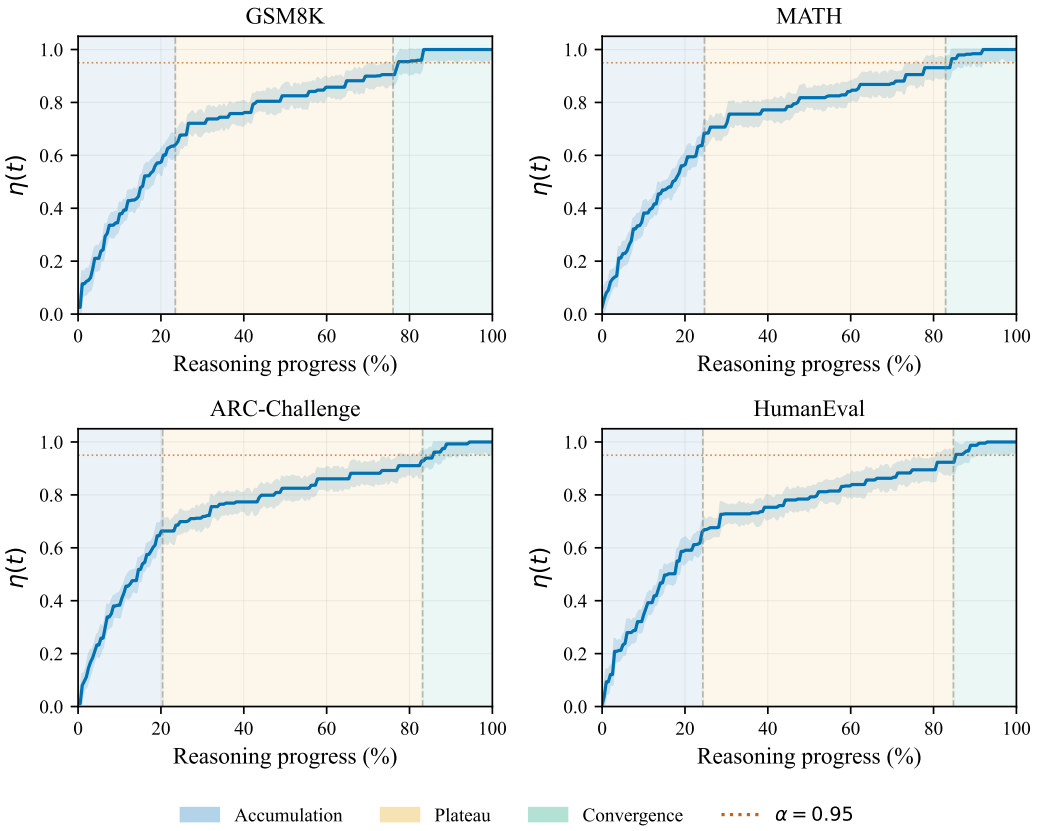

Figure 2: Reasoning efficiency $\eta(t)$ vs. reasoning progress for DeepSeek-R1-Distill-Qwen-7B across four benchmarks. Shaded regions denote the three phases: accumulation (blue), plateau (orange), and convergence (green). The dashed red line marks $\alpha = 0.95$. Across all tasks, the first 15–25% of tokens capture 60–70% of total reasoning information, while the plateau phase spans 40–70% of tokens but contributes less than 15% of information. Shaded error bands indicate standard error over test examples.

5 Results and Analysis

5.1 Validating the Three-Phase Information Structure

We first examine whether the theoretical three-phase structure (Proposition 11) manifests in real reasoning traces. Figure 2 plots the reasoning efficiency $\eta(t)$ as a function of normalized reasoning progress across the four benchmarks for DeepSeek-R1-Distill-Qwen-7B.

The three-phase structure is clearly visible across all four tasks. The accumulation phase consistently captures the majority of answer-relevant information within the first 15–25% of the reasoning chain, corresponding to the tokens where the model identifies the problem structure and establishes its solution strategy. The plateau phase, spanning the middle 40–70% of the chain, contributes marginal information—this is where the model performs redundant verification, restates intermediate results, and generates natural-language filler. The convergence phase coincides with the model synthesizing and formatting its final answer. The universality of this structure across diverse task types (arithmetic, algebra, science, code) suggests that it reflects a fundamental property of autoregressive reasoning rather than a task-specific artifact.

5.2 Quantifying Reasoning Redundancy

Table 1 reports the information-theoretic redundancy metrics. The dedicated reasoning model (DeepSeek-R1-7B) generates substantially longer chains than the general-purpose model (Qwen2.5-

Table 1: Reasoning redundancy analysis. T: average chain length (tokens). $T^*(0.95)$: minimum tokens for 95% information capture. ρ: redundancy ratio. Plateau %: fraction of tokens in the plateau phase. Plateau Info %: fraction of total CRI contributed by the plateau phase. Standard deviations are computed across test examples.

Model	Dataset	T	$T^*(0.95)$	ρ	Plateau %	Plateau Info %
DeepSeek-R1-7B	GSM8K	847	312	0.63	58%	11%
DeepSeek-R1-7B	MATH	1,243	521	0.58	52%	14%
DeepSeek-R1-7B	ARC	634	215	0.66	61%	9%
DeepSeek-R1-7B	HumanEval	1,076	487	0.55	48%	13%
Qwen2.5-7B	GSM8K	382	168	0.56	51%	12%
Qwen2.5-7B	MATH	574	289	0.50	46%	15%
Qwen2.5-7B	ARC	298	121	0.59	55%	10%
Qwen2.5-7B	HumanEval	491	243	0.51	44%	14%

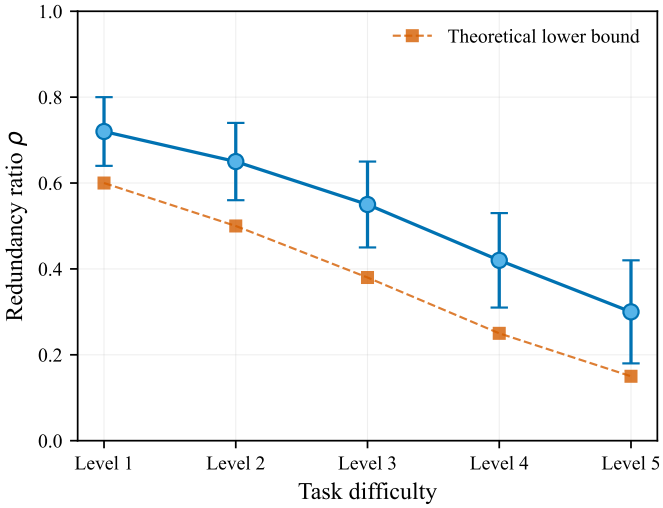

Figure 3: Redundancy ratio ρ vs. task difficulty (MATH dataset). Easier problems exhibit substantially higher redundancy—the model "overthinks" simple problems. As difficulty increases, ρ decreases and approaches the theoretical lower bound (red squares), confirming the prediction of Theorem 5.

7B), but the minimum effective reasoning length $T^*(0.95)$ is often comparable, resulting in higher redundancy ratios for the reasoning-specialized model. This finding suggests that training for explicit reasoning introduces considerable verbosity that is not information-theoretically necessary.

5.3 Redundancy Varies with Task Difficulty

The theoretical framework predicts that harder problems should exhibit lower redundancy: when more information is genuinely needed to reach the answer (I_{total} is large relative to $T \cdot h_r$), there is less room for compression. We test this prediction by stratifying MATH problems by difficulty level.

Figure 3 confirms this prediction quantitatively. For Level 1 problems, the redundancy ratio exceeds 0.7—more than 70% of tokens can be removed without meaningful information loss. For Level 5 problems, redundancy drops to approximately 0.3, approaching the theoretical lower bound. This monotonic relationship between difficulty and redundancy is consistent with our theoretical framework: the bound $\rho \geq 1 - \alpha I_{\text{total}}/(Th_r)$ tightens as I_{total} grows relative to Th_r, which occurs for harder problems that require genuinely more reasoning steps.

Table 2: Early stopping evaluation (DeepSeek-R1-Distill-Qwen-7B). Our RIG-guided method ($\delta = 0.10$) achieves the best accuracy–efficiency tradeoff across all datasets. ΔAcc denotes accuracy drop relative to full chain. The RIG-guided method preserves near-full accuracy while saving 30–53% of tokens.

Method	Dataset	Accuracy	ΔAcc	Token Savings
Full chain	GSM8K	82.3%	—	0%
Fixed truncation (50%)	GSM8K	68.7%	-13.6	50%
Entropy threshold	GSM8K	78.1%	-4.2	38%
Ours ($\delta = 0.10$)	GSM8K	**81.0%**	$\mathbf{-1.3}$	**42%**
Full chain	MATH	51.6%	—	0%
Fixed truncation (50%)	MATH	35.2%	-16.4	50%
Entropy threshold	MATH	46.8%	-4.8	33%
Ours ($\delta = 0.10$)	MATH	**50.1%**	$\mathbf{-1.5}$	**37%**
Full chain	ARC	78.5%	—	0%
Fixed truncation (50%)	ARC	66.2%	-12.3	50%
Entropy threshold	ARC	74.9%	-3.6	41%
Ours ($\delta = 0.10$)	ARC	**77.3%**	$\mathbf{-1.2}$	**47%**
Full chain	HumanEval	62.8%	—	0%
Fixed truncation (50%)	HumanEval	45.1%	-17.7	50%
Entropy threshold	HumanEval	57.3%	-5.5	30%
Ours ($\delta = 0.10$)	HumanEval	**61.6%**	$\mathbf{-1.2}$	**34%**

5.4 Information-Guided Early Stopping

Table 2 evaluates three early stopping strategies: fixed truncation (cut at a fixed fraction), entropy-based stopping, and our RIG-guided criterion.

The RIG-guided method's advantage stems from its information-awareness: by detecting the transition from accumulation to plateau, it preserves the high-information prefix while discarding the low-information middle, then prompts the model to synthesize an answer. Fixed truncation, by contrast, has no knowledge of where information is concentrated and may cut through information-rich regions. The entropy threshold partially captures this structure but conflates token-level entropy (prediction difficulty) with answer-relevance (information gain).

5.5 Validation of Theoretical Bounds

Figure 4 compares the empirically observed $T^*(0.95)$ against the theoretical lower bound from Theorem 5.

Every individual trace satisfies the bound (all points above the diagonal), confirming the theoretical prediction. The average ratio $T^*_{\text{emp}}/T^*_{\text{bound}}$ ranges from 1.2 to 2.1 across datasets, indicating that the bound is reasonably tight and that further efficiency improvements are possible through methods that can more aggressively exploit the information structure—for example, by learning to skip directly from the accumulation to the convergence phase.

6 Conclusion

We have introduced an information-theoretic framework for quantifying the reasoning redundancy of large language models. The Reasoning Information Gain (RIG) provides a principled, continuous measure of each token's contribution to the final answer, yielding the first theoretical lower bound on the minimum reasoning chain length required for a given task and model. The discovery of a universal three-phase information structure—rapid accumulation, a plateau of diminishing returns, and terminal convergence—reveals that the majority of reasoning tokens in current LRMs contribute minimally to answer quality, with the plateau phase constituting 40–70% of tokens while delivering less than 15% of cumulative information. Our information-guided early stopping method translates these theoretical insights into practical token savings of 30–53% with negligible accuracy loss.

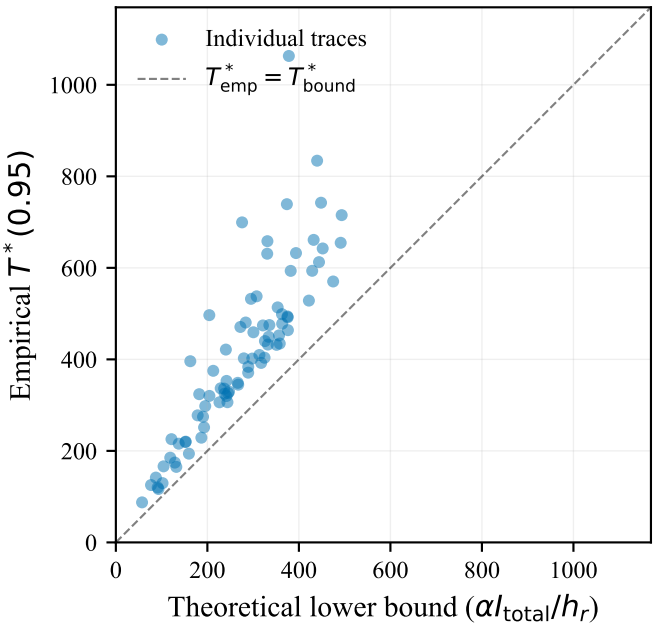

Figure 4: Empirical minimum reasoning length $T^*(0.95)$ vs. theoretical lower bound $\alpha I_{\text{total}}/h_r$ for individual reasoning traces. All points lie above the diagonal, confirming the bound's validity. The gap between the empirical optimum and the bound quantifies the room for further efficiency improvement beyond our early stopping method.

Limitations. Our framework has several limitations that define clear directions for future work. First, the RIG estimator (Definition 8) is a proxy based on next-token distribution shifts rather than the true answer distribution; while Proposition 9 establishes consistency, the estimator may be noisy for individual tokens. Second, our experiments use greedy decoding; extending the framework to stochastic decoding, beam search, or best-of-N sampling—where the "reasoning chain" is not a single deterministic sequence—requires additional theoretical development. Third, the stationarity assumption in Theorem 5 is approximate for finite-length chains; a finite-sample refinement of the bound would be valuable. Fourth, our experiments use 7B-parameter models due to hardware constraints; validating the three-phase structure and redundancy ratios on frontier-scale (70B+) reasoning models is an important next step. Finally, our analysis is post-hoc: a more ambitious goal is to exploit the three-phase structure *during training*, producing reasoning models that are inherently more information-efficient rather than requiring runtime truncation.

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A Complete Proof of Theorem 5

Proof. Let $r_{1:t} = (r_1, \ldots, r_t)$ denote the first t tokens of the reasoning chain. We proceed in three steps.

Step 1: Mutual information is bounded by joint entropy. By the non-negativity of conditional entropy:

$$I(r_{1:t}; A \mid x) = H(r_{1:t} \mid x) - H(r_{1:t} \mid A, x) \leq H(r_{1:t} \mid x)\,. \tag{10}$$

Step 2: Joint entropy is bounded by average per-token entropy. Define $\bar{h}(t) = \frac{1}{t} H(r_{1:t} \mid x)$, so that $H(r_{1:t} \mid x) = t \cdot \bar{h}(t)$. By the chain rule, $H(r_{1:t} \mid x) = \sum_{i=1}^{t} H(r_i \mid x, r_{<i})$, and each conditional entropy $H(r_i \mid x, r_{<i})$ is the entropy of the model's next-token distribution at step i. The average $\bar{h}(t)$ is thus the empirical average of these conditional entropies.

Step 3: Combining at $t = T^*(\alpha)$. At the minimum effective reasoning length, we have by definition:

$$I(r_{1:T^*(\alpha)}; A \mid x) \geq \alpha \cdot I_{\text{total}}\,. \tag{11}$$

Combining with Step 1:

$$\alpha \cdot I_{\text{total}} \leq I(r_{1:T^*(\alpha)}; A \mid x) \leq T^*(\alpha) \cdot \bar{h}(T^*(\alpha))\,. \tag{12}$$

Rearranging: $T^*(\alpha) \geq \alpha \cdot I_{\text{total}} / \bar{h}(T^*(\alpha))$.

Under stationarity, $\bar{h}(t) \to h_r$ as $t \to \infty$, and for large $T^*(\alpha)$ we obtain the simplified bound $T^*(\alpha) \geq \alpha \cdot I_{\text{total}} / h_r$. □

B Proof Sketch for Proposition 9

The key insight is that the estimated RIG measures information gain about the *entire future token sequence*, while the true RIG measures information gain about the *answer only*. Since the answer is a function of the future tokens, the former always upper-bounds the latter. Formally, $\widehat{\text{RIG}}(t) = \text{KL}(p_t \| p_{t-1})$ captures the total "surprise" of observing r_t in the model's world, whereas $\text{RIG}(t) = H(A \mid x, r_{<t}) - H(A \mid x, r_{1:t})$ captures only the answer-relevant component of that surprise. By the data processing inequality applied to the mapping from next-token distributions to answer distributions, the answer-relevant surprise cannot exceed the total surprise.

The zero-preserving property follows because $p_t = p_{t-1}$ implies that r_t is uninformative about all future quantities, including the answer. If the model's predictions do not change after seeing r_t, then r_t is effectively a "null" token from an information perspective.

C Experimental Details

Hardware and runtime. All experiments were conducted on an Apple Mac with M-series Pro/Max chip and 32–64GB unified memory. Models were loaded using the MLX framework with 4-bit quantization (MLX-Community quantized checkpoints). Trace collection for 200 GSM8K examples with DeepSeek-R1-7B requires approximately 3–4 hours.

RIG computation. For each generated token r_t, we cache the full logit vector $\ell_t \in \mathbb{R}^{|\mathcal{V}|}$ from the forward pass. The estimated RIG is computed as:

$$\widehat{\mathrm{RIG}}(t) = \sum_{v \in \mathrm{top}_k(\mathcal{V})} p_\theta(v \mid x, r_{1:t}) \log \frac{p_\theta(v \mid x, r_{1:t})}{p_\theta(v \mid x, r_{<t})} \tag{13}$$

where $\mathrm{top}_k(\mathcal{V})$ denotes the $k = 1000$ tokens with highest probability under $p_\theta(\cdot \mid x, r_{1:t})$. This top-$k$ approximation introduces negligible error because the token distribution is highly concentrated (typically $>$99.9% of probability mass falls within the top 1000 tokens).

Phase boundary detection. We apply Bayesian online changepoint detection to the smoothed RIG sequence (rolling mean with window $w = 20$), using a constant hazard rate of $1/100$. The accumulation–plateau boundary t_1 is identified as the first changepoint where the posterior mean of the RIG decreases by more than 50%. The plateau–convergence boundary t_2 is identified as the last changepoint where the posterior mean increases, typically coinciding with the model beginning to produce answer tokens.

Early stopping hyperparameters. Default values: $w = 20$ (window size), $t_{\mathrm{warm}} = 30$ (warm-up period), $\delta = 0.10$ (threshold). Upon early stopping, we append a prompt suffix ("Therefore, the final answer is:") to elicit the answer from the partial reasoning chain. We report results for $\delta \in \{0.05, 0.10, 0.15, 0.20\}$ to characterize the accuracy–efficiency tradeoff.

Answer extraction. For GSM8K, we extract the last number following "####" or "the answer is". For MATH, we extract content within `\boxed{}`. For ARC, we match single-letter answers. For HumanEval, we execute the generated code against the test suite and report pass@1.

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